Do Now · retrieval

Lesson 125 · Assembly Language and the Little Man Computer

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the purpose of a function return value?

Q2

After pushing A then B onto a stack, which is popped first?

Q3

What is the output of a full adder with A=1, B=1 and carry-in=0?

Q4

Why is the program counter important in a stored-program computer?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Mailboxes — Instructions and numeric data occupy addressed memory mailboxes. DAT reserves a data location. Programs execute the instructions addressed by the program counter.

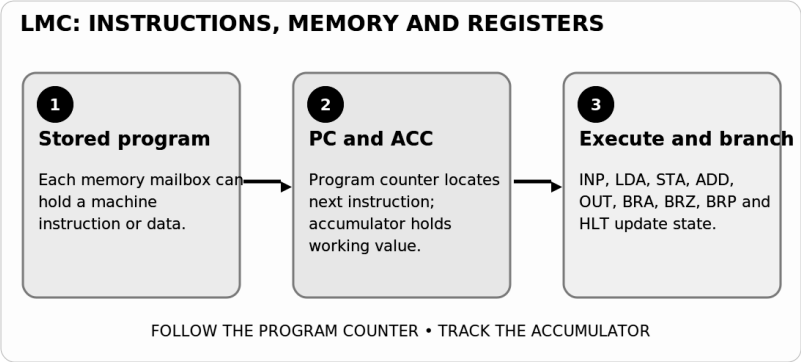
Core idea 2

Accumulator and PC — INP reads into ACC; LDA reads memory into ACC; STA writes ACC to memory. The PC normally advances after a fetch, unless a branch selects a new address.

Core idea 3

Branches and stopping — ADD and SUB update ACC; OUT prints ACC; BRA always branches; BRZ tests for zero; BRP tests a non-negative result in standard LMC; HLT ends the program.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

LMC — Educational model of a stored-program accumulator computer.

accumulator — Register holding current working data and arithmetic results.

program counter — Register holding address of the next instruction to fetch.

mnemonic — Symbolic name representing a machine instruction.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

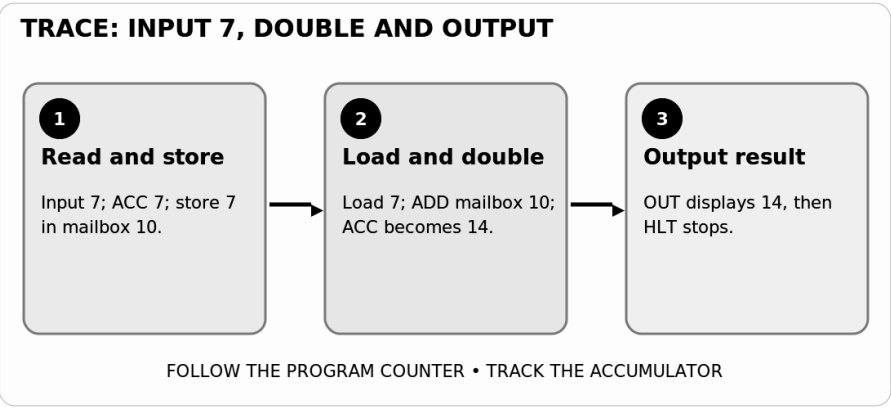
Problem

Input is 7. Trace this program, including the accumulator, data mailbox and output.

Follow the worked method

1. Track each change — 00 INP: ACC = 7. — 01 STA 10: mailbox 10 becomes 7; ACC remains 7. — 02 LDA 10: ACC = 7. — 03 ADD 10: ACC = 14. — 04 OUT: output is 14 . — 05 HLT: stop. Data mailbox 10 is separate from instruction addresses 00–05.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the registers
What are ACC and mailbox 20 after the first two instructions?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain what the PC, accumulator and memory mailboxes represent in LMC. [6]

Question 2

Trace INP; STA 30; LDA 30; ADD 31; OUT; HLT, with input 5 and mailbox 31 containing 3. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Write LMC instructions to read an integer and output it, using a BRZ branch when zero. Explain how you would test both paths. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In LMC, STA n loads the accumulator with the contents of mailbox n.

Use key vocabulary from memory

1. Explain LMC in your own words.

2. Explain accumulator in your own words.

3. Explain program counter in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.